The examination is being carried out on the **following application documents**

**Description, Pages**

1-14 as published

**Claims, Numbers**

1-14 filed with entry into the regional phase before the EPO

**Drawings, Sheets**

1/4-4/4 as published

## Cited documents

1 Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure.

D1 US 2018/177436 A1 (CHANG ANDREW ROBERT [US] ET AL) 28 June 2018 (2018-06-28)

D2 JP 2020 120807 A (INSTITUTE PHYSICAL & CHEMICAL RES; LIFE SCIENCE INST INC) 13 August 2020 (2020-08-13)

## Disclosure (Art. 83 EPC)

2 Claim 1 refers to "classifying the trajectory data into variables of interest". Said wording is represented as step 306 of figure 3 and the corresponding description in par. [0055]. In step 306, key features derived from academic publications (par. [0049]) and statistically established outcomes (par. [0050]) are used in step 306 to classify the trajectory data somehow collected with biomechanical sensors in step 302. However, the application does not enable a person skilled in the art to carry out said classification step because it is not apparent which classes are produced and how these classes are further used in steps 308,. 310 and 312 of figure 3. Even a broader understanding of the classification in the context of a taxonomy does not enable a person skilled in the art to carry out this feature (Art. 83 EPC).

## Novelty (Art. 54 EPC) and Inventive Step (Art. 56 EPC)

3 Document D1, which is considered as closest prior art, discloses all features of claim 1 of the present application:

A method for generating a physiological assessment of a user from biomechanical data gathered from the user, the method comprising the steps of:

the overall goal of the method is best understood from par. [0037] of the application;

D1 discloses such a method, see figure of D1;

acquiring biomechanical data of the user;

see step S110 of D1;

generating trajectory data of the user using the acquired biomechanical data;

trajectory is understood as the "user activity graph" in step S140 of figure 4 and therefore anticipated;

classifying the trajectory data into variables of interest data according to a set of key biomechanical features;

as far as understood, this step is anticipated by D1, pars. [0070], [0083], [0096], or [0112];

rejecting variables of interest data that do not fit a pre-determined acceptance criteria;

extracting key features of the non-rejected variables of interest data;

the method of D1 only uses a subset of attributes, therefore applying filters and rejecting not-used variables, see

pars. [0070], reading "generating a set of stride-based biomechanical signals comprising segmenting kinematic data by steps" and further "Requiring two or more conditions to be satisfied to count segments may improve accuracy of the segmentation"

par. [0083], reading "detecting segments of the sensor data with asymmetric gait dynamics";

par. [0096], reading "detecting segments of sensor data indicative of shuffling gait patterns can include detecting vertical step displacements of the right and/or left steps and classifying the gait as shuffling when vertical step displacements satisfy a shuffle condition" or

par. [0112] reading "when a tremor metric satisfies some tremor threshold condition, the trigger condition of block may be satisfied triggering a response";

for the extraction read also par. [0066], reading "There are a number of indicators that lead to higher risk of falling that can be extracted through biomechanical analysis";

calculating a current cumulative risk data based on the extracted key features, a user profile covariate data and the variables of interest data; and

see D1, pars. [0104]-[0106];

generating the physiological assessment of the user using the current cumulative risk data.

see D1, par. [0108], reading " if a user exhibits sudden changes in movement behavior or walking gait, or exhibits abnormal gait such as a limp or stumble, the device will detect these events and increase the risk profile of the individual. The system can take in multiple inputs, including the ones described above to determine the risk profile of an individual."

3.1 Claim 1 cannot therefore be considered as novel (Art. 54 EPC.

4 The same conclusions can be reached by document D2, using the passages cited in the search report for an equivalent argumentation.

Further claims

5 The same reasoning applies, mutatis mutandis, to the other independent system claim 8 that corresponds to claim 1. Said claims are therefore also neither considered as novel.

6 Dependent claims 2-7 and 9-14 do not appear to contain any additional features which, in combination with the features of any claim to which they refer, meet the requirements of the EPC with respect to novelty (Art. 54 EPC), the reasons being as follows:

- Claims 2 and 9 refer to:

- sorting and labeling the trajectory data into discrete segments and is disclosed by par. [0070] of D1, and
- filtering the discrete segments of trajectory data so to reduce individual frames and remove noise and is disclosed by par. [0070] of D1 relating to improving the accuracy, and
- the use of filtered discrete segments of trajectory data is disclosed by par. [0070] of D1 referring to the counting of segments.

- Further formatting and simplifying the risk data for presentation to the user as in claims 3 and 10 is disclosed by D1, pars. [0050]-[0051].
- Adding biomechanically derived information to the assessment of the user as in claims 4 and 11 is understood as the data received from the sensors, see D1, pars. [0035], [0052], [0054], [0108], and [0132].
- Using combination of curated data from a reference database and an outcome database as in claims 5 and 12 is disclosed as the calibration in par. [0064] of D1.
- Using biomechanical sensors positioned on the user as in claim 6 are disclosed by D1, par. [0027], reading "sensors can be worn ...".
- Using inertial and angular sensors positioned on a the lower-body orthotic device worn by the user as in claims 7 and 13-14 is disclosed by D1, pars. [0027]-[0031] disclosing the use of an inertial measurement unit and the detection of a relative angular orientation.

## Conclusion

7 The present set of claims does not comply with the provisions of the EPC.

8 Should the applicant regard some particular matter as patentable, the Applicant is requested to redraft the current version of the claims so that non-obvious subject matter is claimed. To meet the requirements of Rule 42(1)(b) EPC, the prior art documents should be identified in the description and the relevant background art disclosed therein should be briefly discussed.

8.1 In order the show the compliance with the requirements of Art.123(2) EPC, and in accordance with Rule 137(4) EPC, the applicant is kindly asked to indicate the basis of any claim amendment in form of a highlighted version of the claims, with each feature amended over the originally filed claims indicated together with their respective basis in the originally filed application.

- Added features may be indicated by e.g. underlining and providing the respective basis literally thereafter, e.g. in different font or formatting. If different passages are used, it should be reasoned why said passages can be unambiguously combined.
- Deleted features may be indicated in strike-through font. The applicant should indicate the basis for amendments *always with respect to the claims as originally filed*.

The applicant is reminded that for patents and patent applications each word in a sentence, each sentence in a paragraph, and each paragraph in a specification has a meaning.

- Deviating from the terminology as originally filed can only have one purpose, namely to extend the scope of protection beyond the scope of the application as originally filed. The applicant is therefore requested for any future amendments to strictly adhere to the terminology of the application as originally filed.
- Deviating from the text as originally filed can only be admitted in unambiguous re-arrangements for improving legibility.

8.2 In any letter of reply, the applicant is requested to **define the difference between the claimed subject matter and the teaching of the prior art (document D1**) and to demonstrate how this difference in subject matter is not **obvious** with respect to the prior art. Furthermore, the applicant is requested to **define the technical problem** underlying the alleged invention and demonstrate how the **proposed solution as claimed solves this problem**. This should be shown by indicating any advantageous and/or unexpected technical effects resulting from the proposed solution. In particular, the wording of the independent claim should relate to the solution of the problem solved by the present application.